MINI PROJECT III YEAR

NIET Greater Noida
Autonomous Institute
Get Future Ready

Problem Statement –Rainfall Prediction Using Machine Learning

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Introduction

- Rainfall prediction plays a key role in agriculture and economy
- Helps farmers decide crop type and irrigation planning
- Important for flood and drought prediction
- Traditional methods rely on manual and statistical approaches
- Machine Learning uses historical data patterns
- Improves prediction accuracy and efficiency

Problem statement

- Weather patterns are highly dynamic and unpredictable
- Traditional forecasting methods are less accurate
- Manual prediction requires time and expertise
- Farmers face losses due to wrong predictions
- Lack of simple and accessible prediction tools
- Need an automated, intelligent ML-based solution

Objectives & Proposed Solution

- Develop a Machine Learning model for rainfall prediction
- Use historical rainfall dataset for training
- Provide accurate annual rainfall prediction
- Design a simple and user-friendly interface
- Reduce dependency on traditional methods
- Help farmers, researchers and planners

Technologies Used

- Programming Language: Python
- Frontend: HTML, CSS, JavaScript
- Backend: Flask or Django framework
- Libraries: Pandas (data handling), NumPy (numerical operations)
- Scikit-learn for ML model building
- Tools: Google Colab , VS Code

Methodology

- Step 1: Collect historical rainfall data from datasets
- Step 2: Clean and preprocess the data (remove null values)
- Step 3: Perform feature selection and normalization
- Step 4: Train Machine Learning models
- Step 5: Evaluate models using performance metrics
- Step 6: Deploy best model into web application

ML Models & Evaluation Metrics

- Linear Regression for basic prediction
- Decision Tree for rule-based learning
- Random Forest for higher accuracy and stability
- Evaluation Metrics used:
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE) and R^2 Score

Limitation

- Depends on data quality
- Limited dataset reduces accuracy
- Weather is unpredictable
- Model may overfit/underfit
- No real-time data integration
- Region-specific performance
- Requires internet connectivity
- Needs regular updates

RESEARCH AND REFERENCES

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Key References

- FAO – Digital Agriculture Reports (2022–2023)
- World Bank – Climate & Agriculture Reports
- IEEE & Springer – ML-based Rainfall Prediction Studies
- Algorithms Used: Random Forest, SVM, ANN
- Libraries: Scikit-learn, Pandas, NumPy
- Dataset Sources: Public weather & meteorological data